



Influence of nanoclay reinforcement on thermal and mechanical properties of high-density polyethylene and polypropylene blends

K P JEEVAN and R KRISHNA PRASAD* 

Department of Chemical Engineering and Materials Science, Center of Excellence in Advanced Materials and Green Technologies, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Coimbatore 641112, India

*Author for correspondence (r_krishnaprasad@cb.amrita.edu)

MS received 30 January 2025; accepted 16 March 2025

Abstract. This study investigates the effect of nanoclay (NC) and alumina on improving the thermo-mechanical properties of high-density polyethylene (HDPE) and polypropylene (PP) blends, compatibilized with polyethylene-grafted maleic anhydride (pEMAH). The modulus of the sample containing 1.5% and 3% NC are 893 MPa and 786 MPa, respectively. The ratio of the modulus of composites to that of the matrix increase proportionally with an increase in the composition of NC for various micromechanical models studied. The fracture energy release rate for Run 1 and Run 2 start at 2.08 and 2.35 kJ m⁻² before aging, and once aged, they increase to 6.17 and 5.82 kJ m⁻². High heat on tensile samples leads to bonding in the polymer. The bonding makes a polymer firm, prevents bending and increases the tensile strength. The COMSOL models predict the tensile strength of simulated values at 27 MPa for the 1.5% NC-reinforced matrix, which is the same as the experimental tensile value. Maximum mass loss rates show an increasing trend, with heating rates for the samples containing NC. For instance, the polymer blend containing 1.5% NC has peak mass loss rates of 15% at 5°C, 26% at 10°C and 43% at 15°C as the temperature increases. Adding NC particles to the blend improves its temperature resistance. The activation energy found using Horowitz and Metzger plots for HDPE/PP is 113 kJ mol⁻¹, which increases to 141 kJ mol⁻¹ for the 1.5% NC blend.

Keywords. Micromechanical modelling; fracture analysis; COMSOL modelling; thermogravimetric degradation; Horowitz and the Metzger model; Piloyan-Novikova method.

1. Introduction

The polymers exhibit several properties of interest, such as resistance to heat flow, regular arrangement of polymer chains in the crystalline structure and elasticity. The other properties of polymers are the permeability of gases, resistance to abrasion, and the extent of expansion or contraction of the polymer due to heat flow [1]. Polymers consist of macromolecules that repeat continuously to form a polymeric chain. Thermoplastics are long polymer chains held together by intermolecular forces that do not contain any cross-linking bonds. Thermosetting polymers form a three-dimensional structure on heating due to cross-linking between polymer chains [2]. Polypropylene (PP) manufacturing from the polymerizing of propene monomer exhibits excellent resistance to acids, alcohols, esters and ketones, retains mechanical and electrical properties at high-temperature ranges, resists environmental stress cracking and has a high strength for the definite weight of the polymer. PP contains an unbent chain of propylene monomer with molecules arranged in high adaptability. High-density polyethylene (HDPE) exhibits high specific strength, not a heavily branched structure, high chemical resistance,

resistance to low-temperature exposure, and high impact and tear resistance [3–5].

The polymer blends are miscible when interfacial tension among the polymer phases is absent. The interfacial tension between polymers leads to the separation of polymer phases, where coalescence occurs, resulting in weakened mechanical properties [6]. The natural properties of PP can be enhanced by nanofiller additives, which involve a copolymerization process using two different monomer species to form nanocomposites [7]. The introduction of cellulose fillers changes the properties of low-density PP, similarly, copolymerization with HDPE and PP monomers enhances the insulation capacity for polymer composites [8–10]. PP finds application in various fields due to its chemical resistance and fatigue resistance in addition to low density plus electrical insulation properties that it exhibits. The insulation cables are manufactured using PP [11–14].

The static mechanical and thermal properties improved when the sheet moulding process dispersed carbon fibre and glass fibre in PP [15–17]. Graphene nanoplatelets dispersed through injection moulding in PP improved the tensile stress to 33 MPa (from 16 MPa) and enhanced impact strength by 132%. A surface coating approach with

graphene nanoplatelets increased PP tensile strength by 14% with the addition of 1% graphene nanoplatelets for automotive and aerospace manufacturing. The introduction of nanoclay (NC) into PP via melt blending led to more than 5% increase in tensile strength based on several studies [18–21]. The mechanical and dielectric qualities of graphene sheets improved by mixing them with PP due to an extraordinary quantum tunneling effect [22].

Polymer blends face miscibility problems. Polymer miscibility depends on the chemical nature of polymers, which undergo phase separation at the molecular level [23, 24]. The larger molecules in the blend ensure very little mixing entropy; the interfaces are too narrow, affecting the entanglement of different polymers. The immiscibility of polymer blends decreases due to reduced interfacial tension. Adding polyethylene-grafted maleic anhydride (pEMAH) as a compatibilizer improves the cross-linking of the HDPE and PP polymer blend. The compatibilizer reduces the interfacial tension of the polymer matrix and dispersed filler by chemically interacting with both species or having a unique interaction with one of the components and a physical interaction with the other. It also initiates the coalescence process of the dispersed phase [25–27]. Identifying the properties of a polymer blend entails controlling its microstructure, since multiphase behaviour depends on two polymers and the microstructure [28]. The material is not very stiff because of its amphiphilic nature held together with the polymer matrix. The propylene monomers connect to the long chain of PP, which results in the aggregation of HDPE. pEMAH can graft several polyolefins on both ends, which can solve the aggregation. The compatibility improves due to the increased thickness between the interfacial polymer layers. The interfacial interactions between pEMAH and composites of polymers improve its physical properties [29–33].

Anisotropic layered fillers like NC, graphene, nanographite, and carbon nanotubes and isotropic fillers like nanosilica are added to polymer blends to provide high mechanical strength, superior barrier properties, and excellent thermal properties [34, 35]. The polymer blends interact with nanofillers as intercalated (separating into individual layers) or exfoliated (containing nanometer-scale individual layers) composites. The nanofillers dispersed in polymer blends improve the miscibility of polymers, creating nucleation sites and increasing the tensile properties. The enhancement of tensile properties results from changes in the orientation of the polymer blend rather than just the addition of the filler [36]. The interactions existing across the matrix and fillers reduce the crystallinity and increase the yield strength and elastic modulus [37, 38]. Titanium dioxide is used as filler in the preparation of polyvinyl alcohol and polyethylene oxide nanocomposites for nano-electronics devices and sensors [39–41]. Nanofillers with a high aspect ratio (diameter-to-length ratio) can change the properties of polymers by improving mechanical and barrier properties. In this study, NC is added to the HDPE/PP blend

to enhance the thermo-mechanical properties, and the NC fillers are more easily processable than carbon fillers, thereby reducing the brittleness of polymer blends [42].

2. Materials and methods

2.1 Materials

The HDPE and PP polymers were procured from Reliance Industries Ltd. The grade of HDPE procured from Reliance Industries Ltd. was H050MA180 and the grade of PP was H110MA. The supplier for pEMAH and NC were Saveer Matrix Nano Pvt. Ltd. The composition of NC mainly consists of 99% SiO₂ and 1% alumina. HDPE is denser at 0.95 g cm⁻³, has a melting index at 190°C of 20 g per 10 min, and melts at 130°C. The density of PP is 0.9 g cm⁻³, with a melting index at 230°C of 11 g per 10 min, and it melts at 167°C.

2.2 Methods

The master batch of NC dispersed in the HDPE and PP polymer blend consists of 12.5% (weight) of NC prepared using a lab-scale internal mixer. A compression moulding machine converts the clumps produced in the internal mixer into thin films, often broken into small pellets through mechanical force. The twin-screw extruder melts the polymer compounds with the addition of pEMAH. The twin-screw extruder from Aasabi Machinery Ltd. features conical screws with an 18/36 configuration. The dimensions of the extruder was a screw diameter of 25 mm, and the length-to-diameter ratio of the extruder was 30:1. The temperature along the barrel from the feed-to-die zone ranged from 150 to 210°C with a 100-rpm speed of the screw for all experiments. A scrap grinder machine converts the polymer strands into pellets. These polymer pellets were then injection moulded using an injection moulding machine of make from Ferromatik Milacron Ltd., and model Sigma 50T. The injection moulding machine was used to prepare the dumbbell-shaped tensile specimens according to ASTM D-638. The processing temperature of the machine was in the range of 180 to 210°C and the mould temperature was maintained at 50°C.

The central composite design was used to study the impact of several variables on a specific response and the results were analyzed using MINITAB 17 software. The polymer blend containing HDPE (60 parts) to PP (40 parts) formed the base matrix. The NC (low level: 3% and high level: 6%) and pEMAH (low level: 0.5% and high level: 1.5%) were used in the experimental studies. Table 1 provides the composition matrix containing the %NC (X₁) and %pEMAH (X₂) used in diverse experimental runs for preparing the polymer blends. The response variables measured are the tensile stress (Y₁) and Young's modulus (Y₂).

Table 1. Central composite design experimental runs for preparation of HDPE/PP/NC polymer blends.

Run	NC % (X ₁)	pEMAH % (X ₂)
1	1.5	0.5
2	3	0.5
3	3	1.5
4	1.5	1
5	0	0.5
6	1.5	1
7	1.5	1
8	0	1
9	3	1
10	1.5	1
11	1.5	1
12	0	1.5
13	1.5	1.5

The response of the variables measured follows the linear regression model as given below [43].

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 \quad (1)$$

Here Y is the response value measured, β_0 the constant of correlation, β_1 and β_2 the coefficients for linear variables, β_{11} and β_{22} the coefficients for the quadratic interaction of variables, β_{12} the interactive effect of two factors and X_1 and X_2 the input factors.

2.3 Physical characterization techniques

The spectral analysis of polymer blends was performed using the Thermo Fisher Scientific FTIR Nicolet iS10 instrument with potassium bromide pellets in the 500–4000 cm^{-1} range.

The field emission scanning electron microscope (FESEM, ZEISS GEMINI 300) at a 15 kV accelerating voltage, analyzed the surface morphology of polymer composites.

A universal testing machine measures the tensile stress and modulus of elasticity of polymer blends. The ultimate tensile stress values were recorded directly from the testing conducted on the universal testing machine, while the slope of the tensile curves provides Young's modulus values.

Thermogravimetric analysis (TGA) of polymer blends utilized a model SDT Q600 V20.9 build 20, module DSC-TGA standard under nitrogen (100 ml min^{-1}) from 0 to 1000°C at 5°C min^{-1} , 10°C min^{-1} , and 15°C min^{-1} heating rates.

X-ray diffraction (XRD) studies explain the spectral regimes of nanofiller-dispersed polymer blends, which were determined using a K-beta filter at a 5°C per min rate in a

5–50°C range at operating conditions of 40 kilovolts and 40 milliamperes.

3. Results and discussion

3.1 Fourier transform infrared spectroscopy (FTIR) of HDPE/PP/NC polymer blends

Figure 1 shows the spectral variation of polymer blends prepared in Runs 1, 2 and 3. Table 2 provides the FTIR spectral characteristic peak, corresponding functional group, and peak symmetry for polymer blends prepared in Runs 1, 2, and 3.

The FTIR spectra of the polymer blend prepared in Run 1, consisting of 1.5% NC compatibilized with 0.5% pEMAH, confirm peaks at 2914 cm^{-1} as a result of CH (stretch of bonds) and CH₂ (stretching), 1470 cm^{-1} shows the CH₂ (bending), 1371 cm^{-1} provides CH₃ bending stretch, and 720 cm^{-1} shows the CH₂ rocking stretch. The FTIR spectra of Run 1 at 1860 cm^{-1} suggest carbon and oxygen asymmetric stretching, and the wavenumber 1188 cm^{-1} indicates the presence of carbonyl asymmetric ring stretch, which indicates the existence of maleic anhydride and the wavenumber at 415 cm^{-1} shows silicon and aluminium, confirming the presence of NC.

In Run 2 (3% NC and 0.5% pEMAH), the spectral peak at 2936 cm^{-1} indicates the carbon-hydrogen and CH₂ stretching. The wavenumber 1480 cm^{-1} provides the presence of methylene bending, 1360 cm^{-1} indicates methyl (CH₃ bending), 1857 cm^{-1} and 1176 cm^{-1} show the existence of –CO asymmetric stretch assigned to the availability

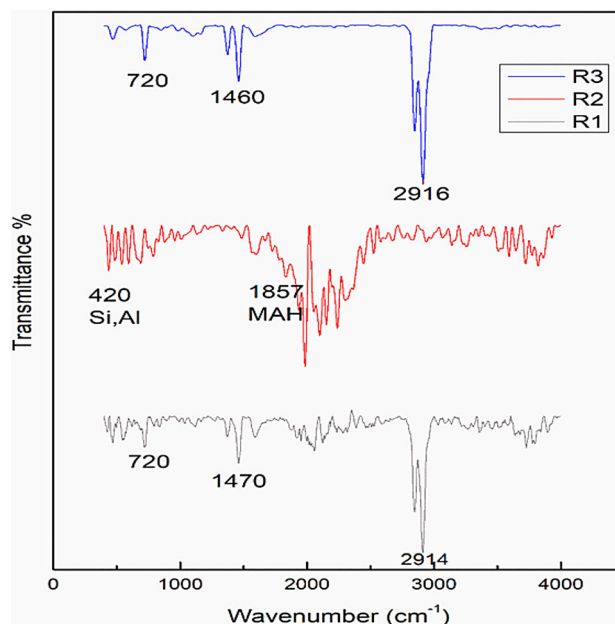
**Figure 1.** FTIR spectra of HDPE/PP/NC blends (Run 1, Run 2 and Run 3).

Table 2. FTIR spectral characteristic peaks of HDPE/PP/NC blends (Run 1, Run 2 and Run 3).

Infrared frequency (cm ⁻¹)	Functional groups	Vibration type	Assigned to the polymer
2980–2800	–CH	Stretching	Polypropylene
2930–2830	–CH ₂	Stretching	High-density polyethylene
1480–1430	–CH ₂	Bending	Polypropylene
1380–1350	–CH ₃	Bending	Polypropylene
1490–1460	–CH ₂	Bending	High-density polyethylene
750–700	–CH ₂	Rocking	High-density polyethylene
470–460	Si, Al	—	Nanoclay
1870–1860	C=O	Asymmetric stretching	Maleic anhydride
1790–1780	C=O	Symmetric stretching	Maleic anhydride
1230–1180	C=O	Asymmetric ring stretching	Maleic anhydride
1060–1050	C=O	Symmetric ring stretching	Maleic anhydride

of pEMAH, and 420 cm⁻¹ indicates silicon and aluminium presence confirming the availability of NC.

The spectral analysis of Run 3 (3% NC and 1.5% pEMAH) confirms the peaks at 2936 cm⁻¹ as a result of CH stretching and methylene (CH₂ stretching), 1355 cm⁻¹ showing methyl (CH₃ bending), 1180 cm⁻¹ revealing carbonyl asymmetric ring stretching assigned to the maleic anhydride, and the wavenumber at 426 cm⁻¹ revealing the existence of silicon and aluminium, confirming the availability of NC.

3.2 Surface morphological analysis

Figure 2 shows the scanning electron microscope (SEM) images of the fractured surface obtained from a tensile

specimen after subjecting it to tensile stress. It shows fractured images of the HDPE/PP blend and Run 2 containing 3% NC at 2 and 4 μm scale bars. The surface dislocation and fracturing are visible in the SEM images of the HDPE/PP and Run 2 specimens at different scale bars. The SEM images of Run 2 at different resolutions show the presence of microcavities responsible for crack propagation. The Run 2 specimen images indicate the uniform spread of NC in the matrix at 3% NC concentration, and cross-linking bonding exists between the filler and the matrix. The Run 2 images demonstrate that the NC disperses within the matrix like a net-like or fibrous structure where NC entwines within the polymer blend. The NC achieves uniform dispersion in the HDPE and PP blend due to large aspect ratio of filler and the incorporation of shear force during melt blending.

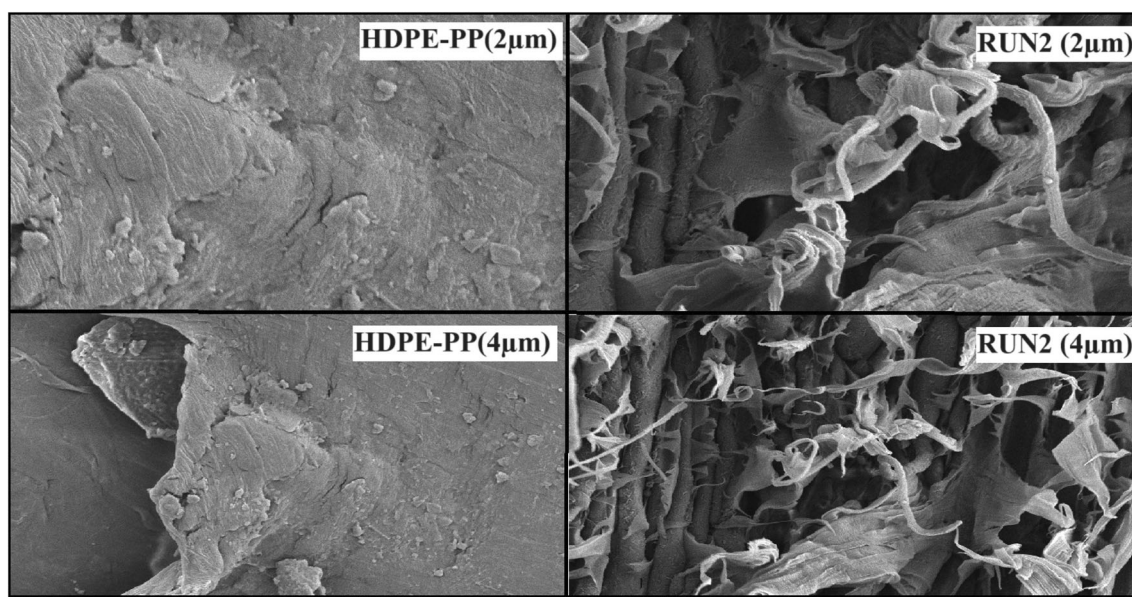
**Figure 2.** SEM images of fractured HDPE/PP and Run 2 polymer blends (Scale bar: 2 and 4 μm).

Table 3. Tensile strength, strain and modulus of polymer blends before aging and after aging.

Samples	Tensile strength (BA) MPa	Tensile strength (AA) MPa	Strain % (BA)	Strain % (AA)	Modulus (BA) MPa	Modulus (AA) MPa
HDPE/PP	26.2	22.0	11.2	3.5	295	266
1	28.7	32.5	15.7	8.4	893	387
2	28.6	33.1	19.5	7.5	786	425
3	28.5	31.3	18.7	6.1	704	391
4	28.6	31.9	17.2	8.9	645	422
5	22.2	30.4	9.2	9.2	778	464
6	28.6	31.9	17.2	8.9	645	422
7	28.6	31.9	17.2	8.9	645	422
8	28.3	30.8	17.2	7.0	728	378
9	28.6	31.9	21.8	8.9	748	476
10	28.6	31.9	17.2	8.9	645	422
11	28.6	31.9	17.2	8.9	645	422
12	28.5	30.4	36.6	7.5	806	426
13	27.7	31.2	40.8	8.2	892	471

3.3 Static mechanical properties of HDPE/PP/NC polymer blends

Table 3 provides the variations in tensile stress and modulus of elasticity of polymer samples before and after aging. The Run 1 (1.5% NC and 0.5% pEMAH) polymer blend provides an ultimate tensile stress value of 28.7 MPa. The NC composition is increased in Run 2 (3% NC and 0.5% pEMAH), which provided 28.6 MPa tensile stress. The NC content increases in Run 3 (3% NC and 1.5% pEMAH), and the resulting polymer blend broke at a stress value of 28.5 MPa. In comparing Runs 8 and 12, having no NC content for an increase in pEMAH content from 1% to 1.5%, the polymer blends improved tensile stress from 28.3 MPa to 28.5 MPa. Increasing NC content material for the identical level of pEMAH addition confirmed no change in tensile stress. The polymer blends from Run 4 and Run 9, wherein NC increases from 1.5% to 3% for a 1% pEMAH addition, provided an ultimate tensile value of 28.6 MPa.

The modulus of the matrix reduces as NC composition increases at the same degree of pEMAH addition. The Run 1 (1.5% NC and 0.5% pEMAH) exhibits a maximum modulus level of 893 MPa, and it is reduced in Run 2 (3% NC and 0.5% pEMAH) to 786 MPa. Run 2 is compared with Run 3 (3% NC and 1.5% pEMAH) showed a reduction in modulus from 786 MPa to 704 MPa. The Run 8 (0% NC and 1% pEMAH) and Run 12 (0% NC and 1.5% pEMAH) are compared for analyzing the increase in pEMAH content, which showed a rise in modulus value from 728 MPa to 806 MPa.

The regression equation for tensile strength (uncoded units) is as follows.

$$Y_1 = 18.3 + 3.79X_1 + 12.38X_2 - 0.303X_1^2 - 3.72X_2^2 - 2.133X_1X_2 \quad (2)$$

The regression equation for the modulus (uncoded unit) is:

$$Y_2 = 1210 + 44.1X_1 - 1105X_2 - 5.2X_1^2 + 571X_2^2 - 36.7X_1X_2 \quad (3)$$

Figure 3 provides the contour plots for ultimate tensile stress and the modulus of polymer blends. From the contour plot drawn for NC% vs. pEMAH%, we can infer that 3% NC and 1% pEMAH polymer blends achieve a maximum tensile strength of over 29 MPa. The modulus value of blends is more than 800 MPa for 1.5% NC and 1.5% pEMAH composition, and the modulus value of 3% NC and 1.5% pEMAH composition of blends is above 750 MPa. The contour plots suggest that 3% NC polymer blends exhibit maximum mechanical properties in the blends.

3.4 Studies on the thermal aging of HDPE/PP/NC polymer blends

The critical stress intensity factor (K_c ($\text{MPa m}^{0.5}$)) represents the fracture toughness of most of the materials. We calculate the value of K_c using the following equation:

$$K_c = \sigma\beta\sqrt{\pi a} \quad (4)$$

here, σ represents the applied stress in MPa, β is a dimensionless correction factor and a represents the crack length in meters.

The following relationship provides the fracture energy release rate, G_c (kJ/m^2).

$$G_c = \frac{K_c^2}{E^*} \quad (5)$$

Here E^* represents the equivalent modulus.

Table 4 displays the critical stress intensity factor and strain energy release rate of polymer blends before and after

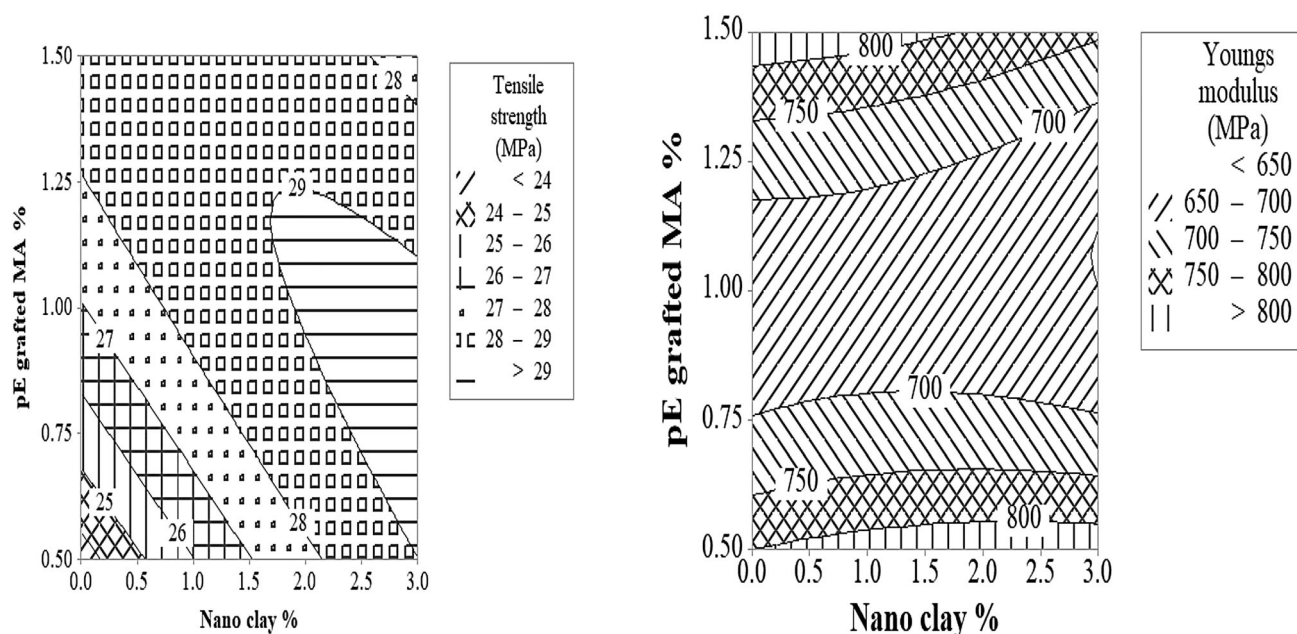


Figure 3. Contour plots for tensile strength and Young's modulus.

aging. The study examined the impact of heat aging on dumbbell tensile specimens under accelerated conditions at 100°C for 144 h. They were left at room temperature for 24 h before being subjected to tensile tests. After exposure to a higher temperature for a definite time, nanofillers in the polymer specimen undergo redistribution within the matrix. It affects the stress distribution within the material and results in enhanced tensile strength and stiffness. The crack length for determining the K_{Ic} values is 0.002 m. The K_{Ic} values of thermally aged samples rise as filler goes up. Before aging, Run 1 and Run 2 have K_{Ic} values of 1.36 and 1.35 MPa m^{0.5}. After aging, they are 1.54 and 1.57 MPa

m^{0.5}. The G_c values for Run 1 and Run 2 start at 2.08 and 2.35 kJ m⁻². Once aged, they increase to 6.17 and 5.82 kJ m⁻². High heat on tensile samples leads to bonding in the polymer. The bonding makes a polymer firm, prevents bending and increases the tensile strength.

3.5 Micromechanical modelling of the elasticity modulus of polymer blends

In this section, experimental moduli variations relate to the theoretical models for NC content of varying compositions

Table 4. The failure load, critical stress intensity factor (K_{Ic}), strain energy release rate (G_c) of polymer blends before and after aging.

Samples	Failure load (BA) N	Failure load (AA) N	K_{Ic} , BA (MPa. m ^{1/2})	K_{Ic} , AA (MPa. m ^{1/2})	G_c , BA (kJ m ⁻²)	G_c , AA (kJ m ⁻²)
HDPE/PP	1020	1250	1.24	1.52	5.26	8.75
1	1120	1270	1.36	1.54	2.08	6.17
2	1100	1290	1.35	1.57	2.35	5.82
3	1010	1220	1.35	1.48	2.60	5.66
4	1100	1240	1.35	1.51	2.86	5.45
5	1110	1190	1.05	1.44	1.43	4.50
6	1100	1240	1.35	1.51	2.86	5.45
7	1100	1240	1.35	1.51	2.86	5.45
8	1100	1200	1.34	1.46	2.48	5.67
9	1100	1240	1.35	1.51	2.47	4.83
10	1100	1240	1.35	1.51	2.86	5.45
11	1100	1240	1.35	1.51	2.86	5.45
12	1110	1190	1.35	1.44	2.27	4.90
13	1080	1220	1.31	1.48	1.94	4.67

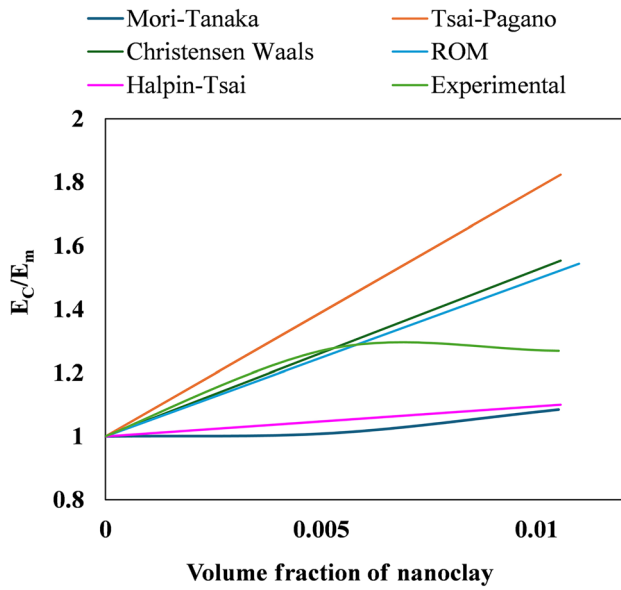


Figure 4. Micromechanical modelling approach of comparison of experimental and theoretical moduli of HDPE/PP/NC polymer blends with varying NC content.

in the polymer blends. Several assumptions considered in the model are perfect alignment, homogeneous dispersion of fillers, and the fillers and matrix exhibit an isotropic

nature. The axial modulus of NC-reinforced polymer blends is determined using the rule of mixtures (ROM) model.

$$E_C = \eta_1 \eta_0 \phi_f E_f + (1 - \phi_f) E_m \quad (6)$$

The model includes a length efficiency factor (η_1) that accounts for the complex, varying alignment of fibres in the base matrix and an orientation factor (η_0) that depends on the dimension of the filler. It also includes the volume fraction of NC (ϕ_f). E_f , E_C and E_m are the values of the modulus of NC, HDPE/PP/NC composite, and HDPE/PP matrix, respectively.

The comparison of modulus of HDPE/PP/NC composite to the HDPE/PP matrix along a parallel direction to the length of fibre's is determined using the Mori-Tanaka model:

$$\frac{E_C}{E_m} = \frac{A}{A + \phi_f [A_1 + 2\gamma_m A_2]} \quad (7)$$

where A_1 , A_2 and A are functions of Eshelby's transformation tensors, and their components depend on the volume fraction of the NC and Poisson's ratio of the HDPE/PP matrix (γ_m). The Poisson ratio of the matrix is 0.42 and the Poisson ratio of the filler is 0.278.

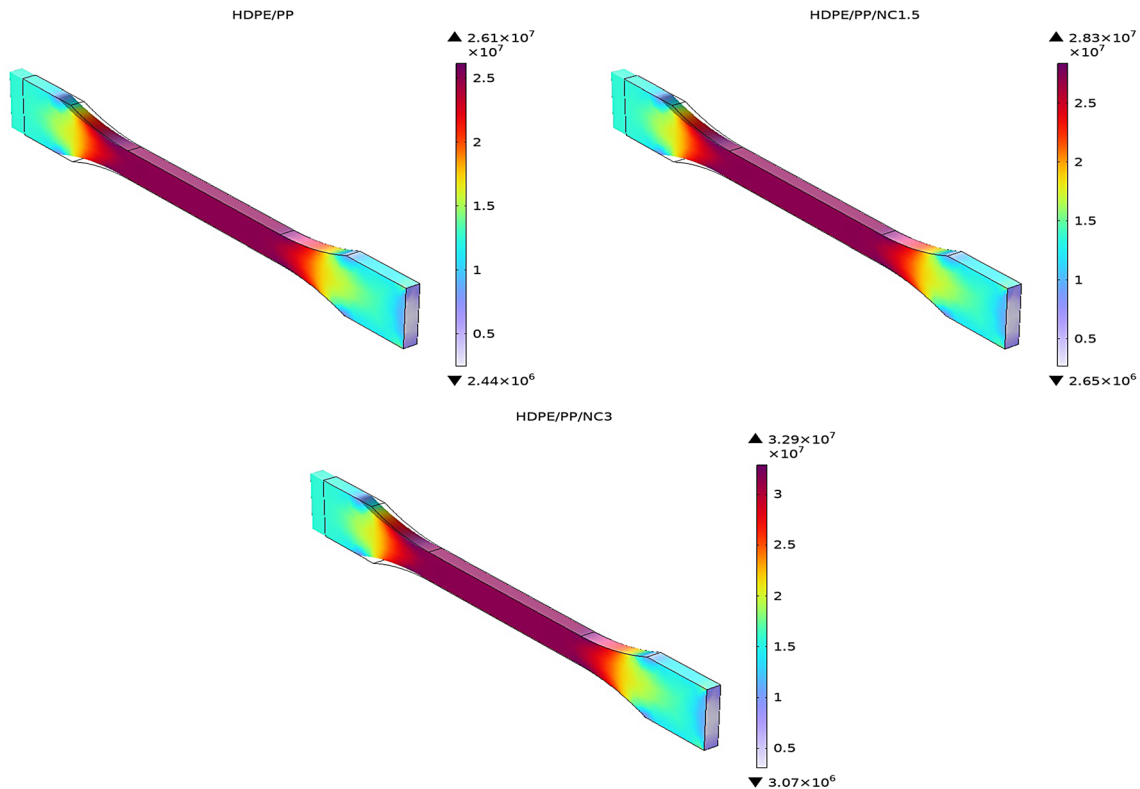


Figure 5. Predictive COMSOL modelling of tensile stress showing contour regions of tensile stress in specimen.

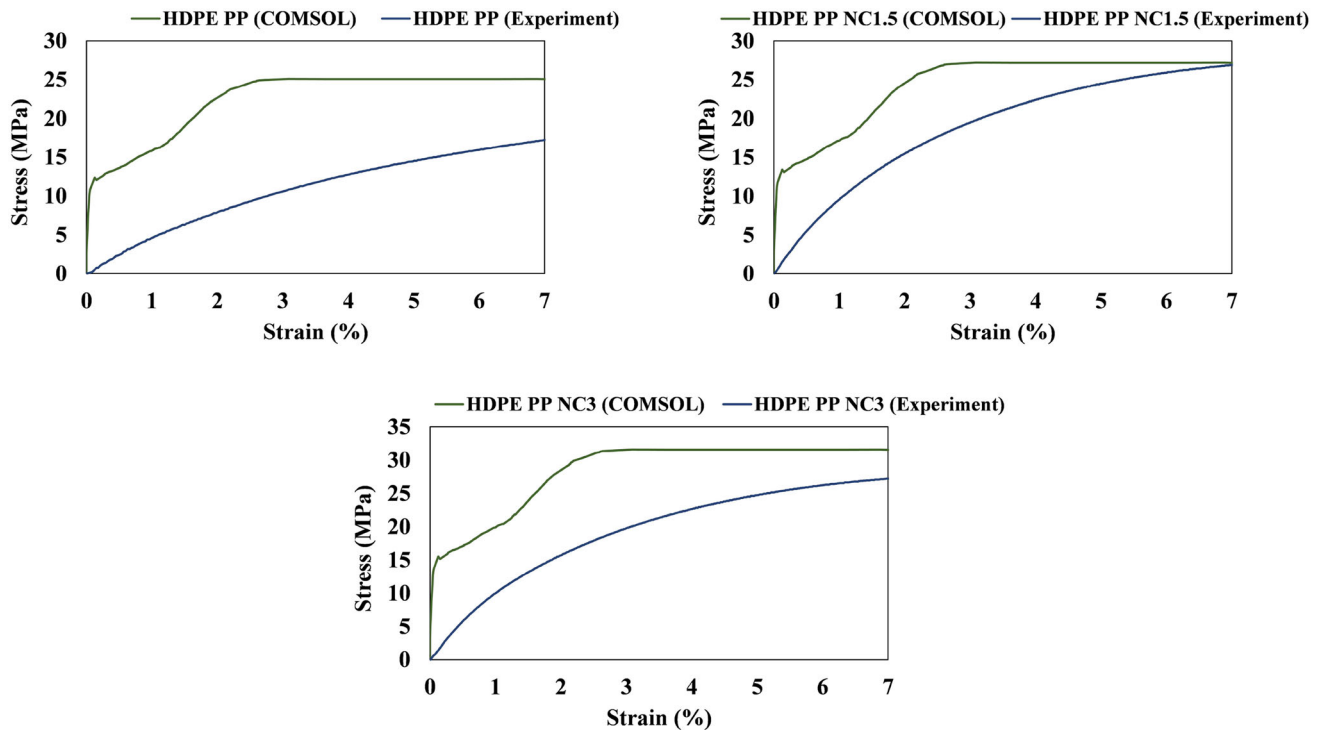


Figure 6. Comparison of tensile stress obtained experimentally with the predicative values attained from COMSOL modelling.

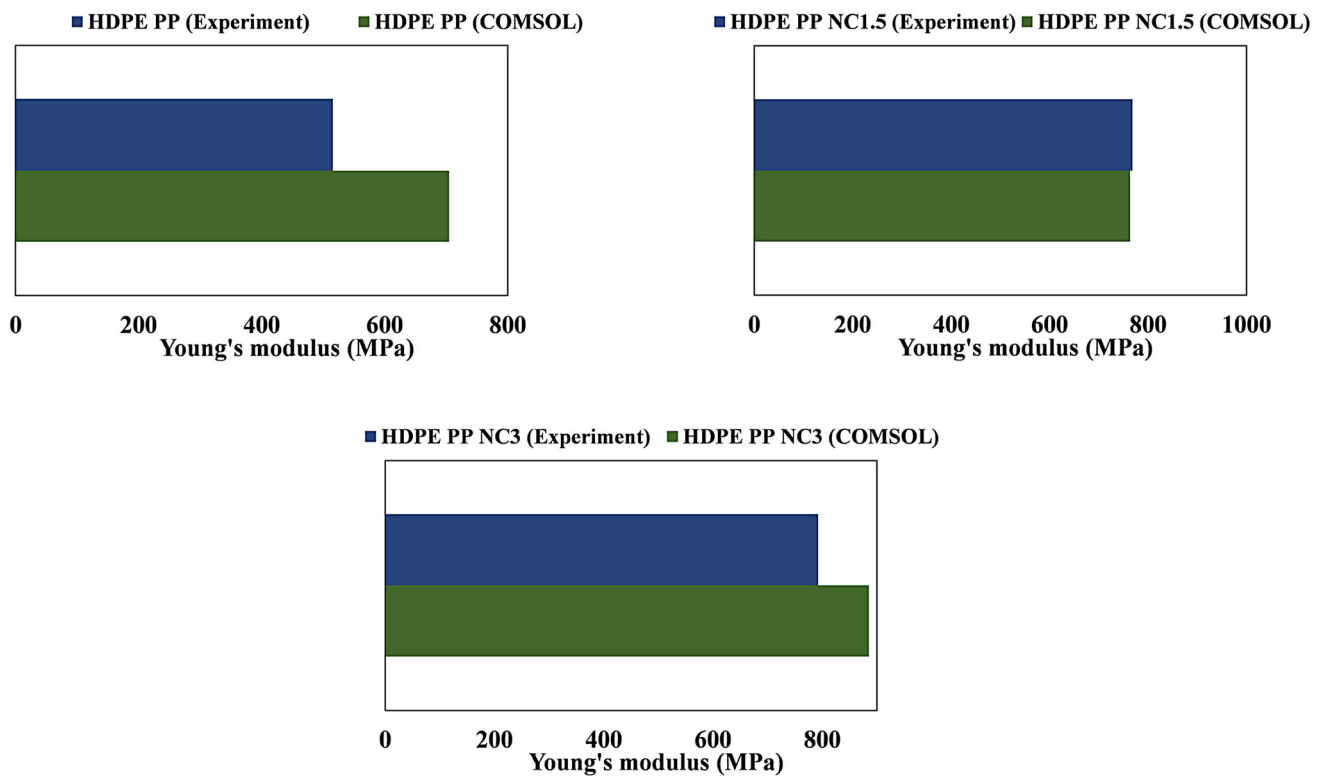


Figure 7. Comparison of Young's modulus obtained experimentally with the predicative values attained using COMSOL modelling.

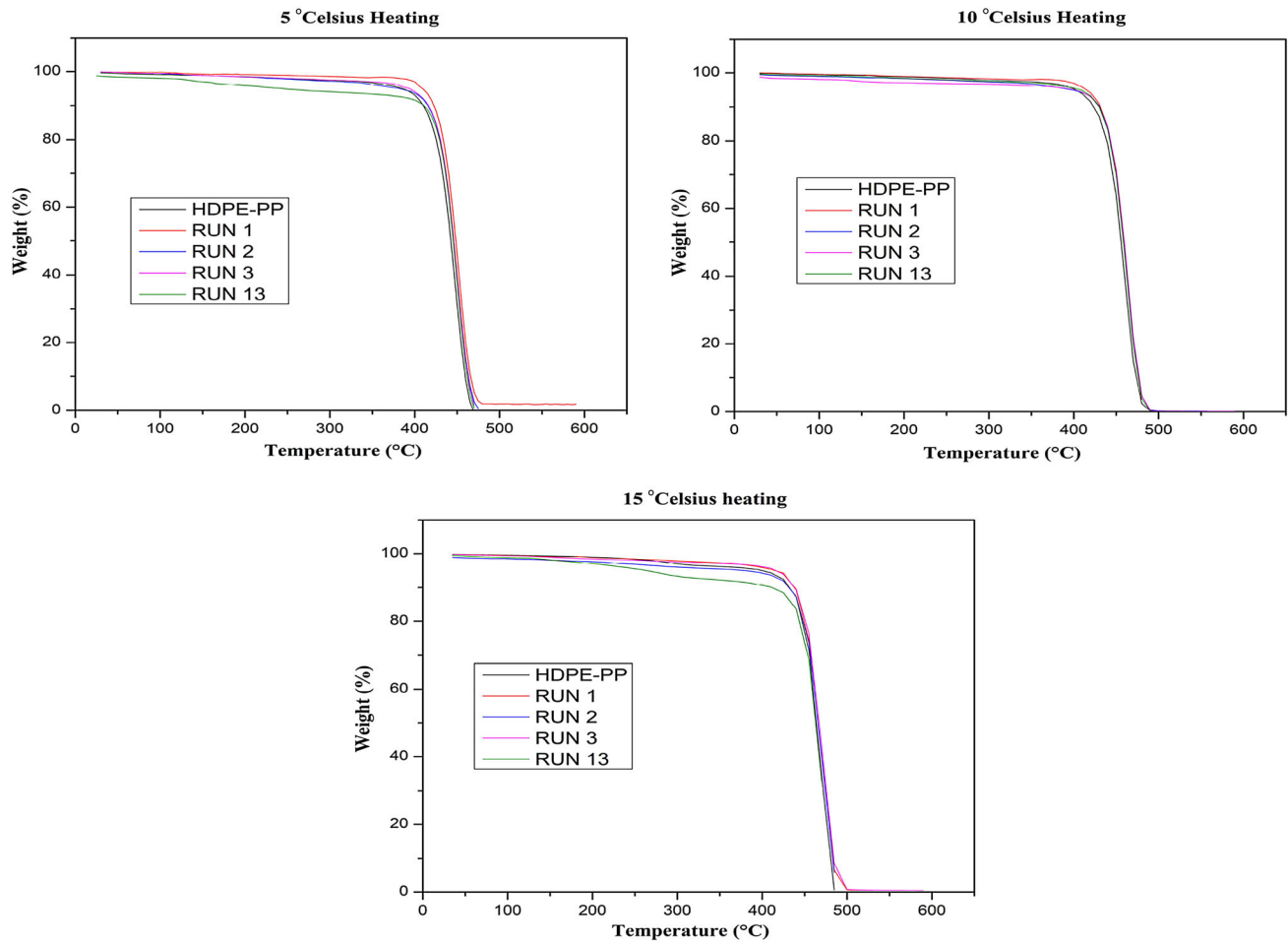


Figure 8. Thermogravimetric decomposition patterns of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends at different heating rates.

The model developed by Halpin-Tsai evaluates the ratio of modulus as follows.

$$\frac{E_c}{E_m} = \frac{1 + \xi \eta \phi_f}{1 - \eta \phi_f} \quad (8)$$

$$\eta = \frac{\frac{E_f}{E_m} - 1}{\frac{E_f}{E_m} + \xi} \quad (9)$$

The correction factor depends on geometry (ξ), which depends on the ratio of sizes in different dimensions or the aspect ratio of filler (a) ($\xi = 2a$). This model does not consider filler aggregation or the variations across the interfacial regions.

The Tsai-Pagano model is a method to find the modulus in HDPE/PP/NC composites. It uses unidirectional aligned fillers, which are explained in the following equations [44].

$$E_c = \frac{3}{8}E_L + \frac{5}{8}E_T \quad (10)$$

$$\frac{E_L}{E_m} = \frac{1 + \xi_L \eta_L \phi_f}{1 - \eta_L \phi_f} \quad (11)$$

$$\frac{E_T}{E_m} = \frac{1 + \xi_T \eta_T \phi_f}{1 - \eta_T \phi_f} \quad (12)$$

In these equations, L represents longitudinal properties, while T represents transverse properties. The evaluations of ξ_L and ξ_T are done such that they are equal to 2a and 2, respectively.

The model suggested by Christensen and Waals indicates a three-dimensional distribution of fillers in the matrix in random distribution [44].

$$E_c = \frac{\phi_f}{6}E_f + [1 + (1 + \gamma_m)\phi_f]E_m \quad (13)$$

Figure 4 shows the E_c/E_m vs. volume fraction of NC obtained through the experimental method and the other five predictive models. The modulus variation of the HDPE/PP/NC composite and the HDPE/PP matrix changes with varying NC content. The values of the modulus ratio evaluated using the Tsai-Pagano model, the Christensen-Waals model, and the ROM model are close to experimental modulus values at a low NC fraction of lower than 0.005 volume fraction and over predict compared to experimental modulus values at a higher NC volume

Table 5. Thermogravimetric decomposition parameters of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends at different heating rates.

Run	T _{onset} (°C)	T ₁₀ (°C)	T ₅₀ (°C)	dT _{max} (°C)	dw/dt (%/min)
5°C/min heating rate					
HDPE/PP	387	410	442	448	13.15
1	393	422	448	452	15.14
2	393	414	446	450	13.36
3	392	415	446	450	14.72
13	397	410	444	450	14.15
10°C/min heating rate					
HDPE/PP	403	423	456	469	26.86
1	405	429	458	463	25.75
2	405	430	459	463	27.85
3	408	430	460	463	27.88
13	409	430	458	462	27.58
15°C/min heating rate					
HDPE/PP	402	432	464	470	44.93
1	418	437	465	470	42.89
2	410	429	466	471	42.03
3	412	440	466	471	42.27
13	416	411	463	470	43.46

fraction. The modulus evaluated using the Halpin-Tsai model and the Mori-Tanaka model under predict compared to experimental modulus values. The modulus ratio of composites to matrix increases proportionally with the volume fraction of NC in all evaluated models.

3.6 Predictive COMSOL modelling of tensile stress and Young's modulus

The tensile specimens were prepared from the injection moulding machine according to the geometry specified in ASTM D-638 standards. The dumbbell specimen dimensions are $165 \times 19 \times 3.2$ mm according to ASTM D-638 standards. Figure 5 illustrates contour regions of tensile stress of specimens (HDPE/PP, 1.5% NC-reinforced matrix, and 3% NC-reinforced matrix). Figure 6 illustrates the variation of stress with strain for experimental and predicted COMSOL tensile stress of specimens containing various NC compositions. Figure 7 compares Young's modulus from experimental results with predicted values from simulation. The stress values for varying strains are higher in predicted results than the experimental results obtained for neat HDPE/PP and NC-reinforced matrix. The HDPE/PP matrix reinforced with 1.5% NC shows a modulus of elasticity obtained from experimental results similar to the predicted modulus. The experimental modulus of the HDPE/PP blend is 516 MPa and the maximum predicted value of the tensile specimen from the simulation is 704 MPa. Maximum stress is observed at the central region

of the tensile specimen, and stress reduces towards the edges of the specimens. The specimen containing 3% NC exhibits a maximum ultimate tensile stress of 30 MPa from predicative results. The stress-strain graphs show that adding NC to the polymer blend improves the ultimate tensile stress. The pure HDPE/PP blend exhibits an ultimate stress of 17 MPa, and adding NC improves the ultimate stress to 27 MPa, as shown in the figures. The experimental and simulated values of ultimate tensile stress of the 1.5% NC-reinforced matrix are similar at 27 MPa, showing that the model can accurately predict the ultimate stress results.

3.7 Thermogravimetric analysis of HDPE/PP/NC polymer blends

The thermogravimetric degradation pattern was calculated for pure HDPE/PP, Runs 1, 2, 3 and 13, which were analyzed at various heating rates of 5, 10 and 15°C per min. Figure 8 provides the decomposition pattern of the specimens analyzed at different heating rates, and the plotted graph contains the weight percentage remaining vs. temperature. Table 5 records the onset degradation temperature (T_{onset}), the recorded temperature at 10% and 50% weight reduction (T₁₀ and T₅₀, respectively), and the maximum temperature (dT_{max}) at the highest rate of mass loss (dw/dt). Figure 9 presents the derivative thermograms of specimens at different heating rates.

The onset degradation temperature at which thermal decomposition starts is lower for HDPE/PP compared to NC-added samples for all heating rates. The HDPE/PP polymer exhibits thermal stability until 403°C for a 15°C per min heating rate; however, adding 1.5% NC to the polymer blend in Run 1 provides more thermal stability until 418°C for a 15°C per min heating rate. The comparison of Runs 1 and 2, containing 1.5% and 3% NC, shows that adding NC content to polymers has a marginal impact on onset temperature for a 5°C and 10°C per min heating rate. But when the rate of heating increases to 15°C per min, Run 1 sample containing 1.5% NC provides a 418°C onset temperature, while Run 2 with 3% NC has an onset temperature of 410°C. The temperature at which the NC-dispersed polymer blends decompose is higher than the individual polymers, indicating higher thermal stability. The temperature at which the HDPE/PP loses half its weight is lower than the thermal degradation temperature of Runs 1 and 2 containing NC. The 50% weight loss temperature (T₅₀) for Runs 1 and 2 shows a rise in degradation temperature for the addition of NC content for 10°C and 15°C per minute heating rates. At 5°C per min heating rates, 1.5% NC-added polymer exhibits a higher degradation temperature (T₅₀) of 448°C compared to the 3% NC sample showing a T₅₀ value of 446°C.

The maximum mass loss rates (dw/dt) increase with the increase in the heating rates in different runs. The HDPE/PP blend shows a peak mass loss rate of 13, 27 and 45% per min for 5, 10 and 15°C per min heating rates, respectively.

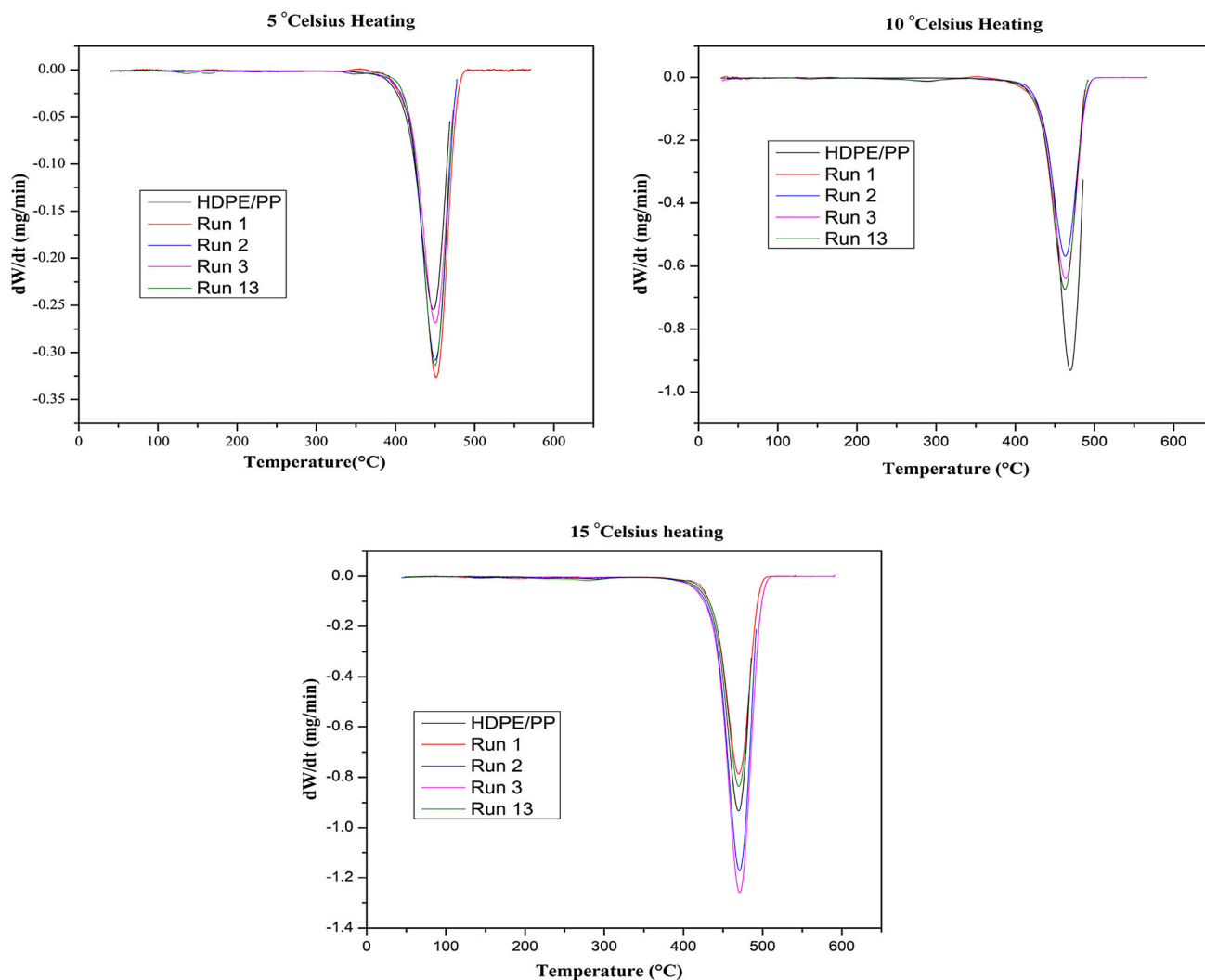


Figure 9. Derivative thermograms of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends at different heating rates.

The blend with 1.5% NC shows peak mass loss rates of 15, 26 and 43% per min at 5, 10 and 15°C per min heating rates, respectively. Comparing Runs 2 and 3, which contain 3% NC and 0.5% and 1.5% pEMAH respectively, shows that the temperature at 50% weight loss remains unchanged despite the increase in compatibilizer content. Adding clay to the blends blocks heat flow and improves the temperature resistance of the blends. Adding clay shifts the temperature at which thermal degradation starts in the polymer blend to higher regions. The layers of silicates absorb heat and accelerate the thermal degradation process. Clay loading into the blends improves the resistance to temperature rise in blends. The degree of thermodynamic changes occurring within the polymer composite affects the thermal stability of NC polymer blends. Comparing Runs 3 and 4 at a rate of heating of 10°C per min, we can observe that the same 1.5% clay addition in both the samples and an increase in pEMAH composition from 0.5% to 1.5% resulted in a reduction of peak mass loss rate from 33.51 to 29.79% per min.

3.8 Thermogravimetric models for determination of activation energy of polymer blends

Horowitz and the Metzger model calculate the degree of conversion (α) using the following equation.

$$\alpha = \left(\frac{w_T - w_f}{w_O - w_f} \right) \quad (14)$$

where w_O is the initial weight of the polymer sample, w_f is the final weight of the remaining polymer material after thermal degradation, w_T is the weight of the polymer material recorded at temperature T .

The following equation calculates the activation energy (E_a) [45]. The slope of the graph of the reciprocal of the double logarithm of the degree of conversion vs. temperature provides E_a .

$$\ln \left[\ln \left[\left(\frac{w_O - w_f}{w_T - w_f} \right) \right] \right] = \frac{E_a \theta}{RT_{\max}^2} \quad (15)$$

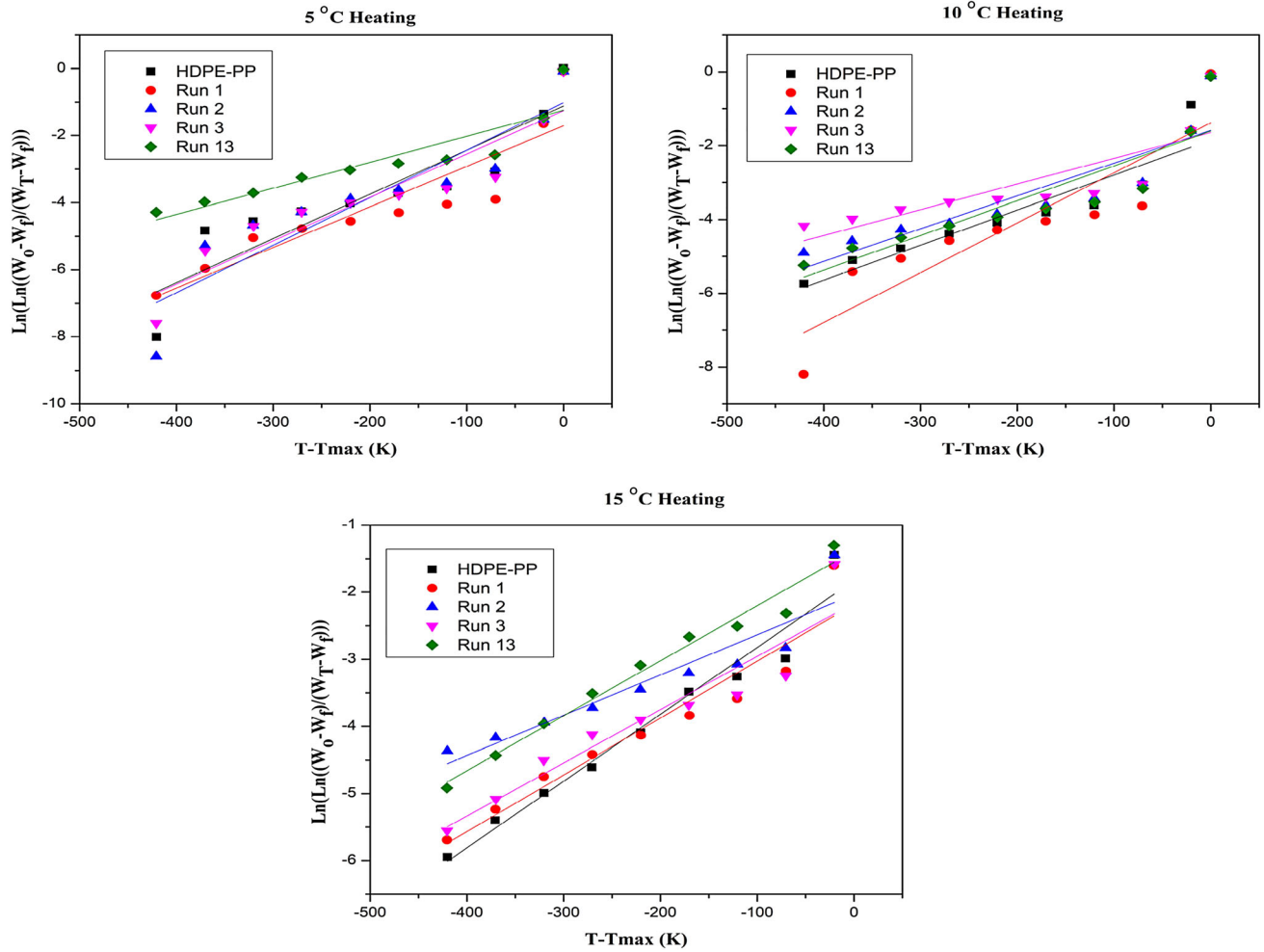


Figure 10. Horowitz and Metzger plots for activation energy determination of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends.

$$\theta = T - T_{\max} \quad (16)$$

where $T_{\max}$ is the temperature corresponding to the peak rate of weight loss, R is the gas constant ($8.314 \text{ J mol}^{-1} \text{ K}$).

Coats and Redfern method of modelling the thermal degradation of polymer composites to calculate E_a with the first order of reaction (n) [46] is as follows:

$$\log \left[1 - \frac{(1-\alpha)^{1-n}}{T^2(1-n)} \right] = \left[\log \left(\frac{AR}{\alpha E_a} \right) \left\{ 1 - \left(\frac{2RT}{E_a} \right) \right\} \right] - \left(\frac{E_a}{2.303RT} \right) \quad (17)$$

The $\log \left[-\log \left(\frac{1-\alpha}{T^2} \right) \right]$ vs. $1000/T$ gives a linear plot with slope $E_a/2.303R$.

Chan model: This method assumed first-order kinetics and used the Arrhenius relation to determine activation energy. They used the Arrhenius plot $\ln(k/w)$ vs. $1/T$, value $-E_a/R$ provides the slope of the linear plot. Here k is the weight

percentage loss per minute at a temperature of T (%/min), w is the weight percentage of polymer remaining at a temperature of T (%).

Broida model: The Broida equation to calculate E_a is as follows:

$$\ln \left[\ln \left(\frac{1}{y} \right) \right] = -E_a + \frac{C}{RT} \quad (18)$$

here y is the weight fraction of the sample remaining at temperature T , plot is between $\ln \left[\ln \left(\frac{1}{y} \right) \right]$ vs. $1/T$.

Piloyan-Novikova model: The equation follows a first-order reaction with a pure kinetic regime [47].

$$\ln \left[\frac{1 - (1-\alpha)1-n}{(1-n)T^2} \right] = \ln \frac{ZR}{\beta E_a} - \frac{E}{RT} \quad (19)$$

$$Z = \frac{kT_m}{h} \exp \left(\frac{\Delta S}{R} \right) \quad (20)$$

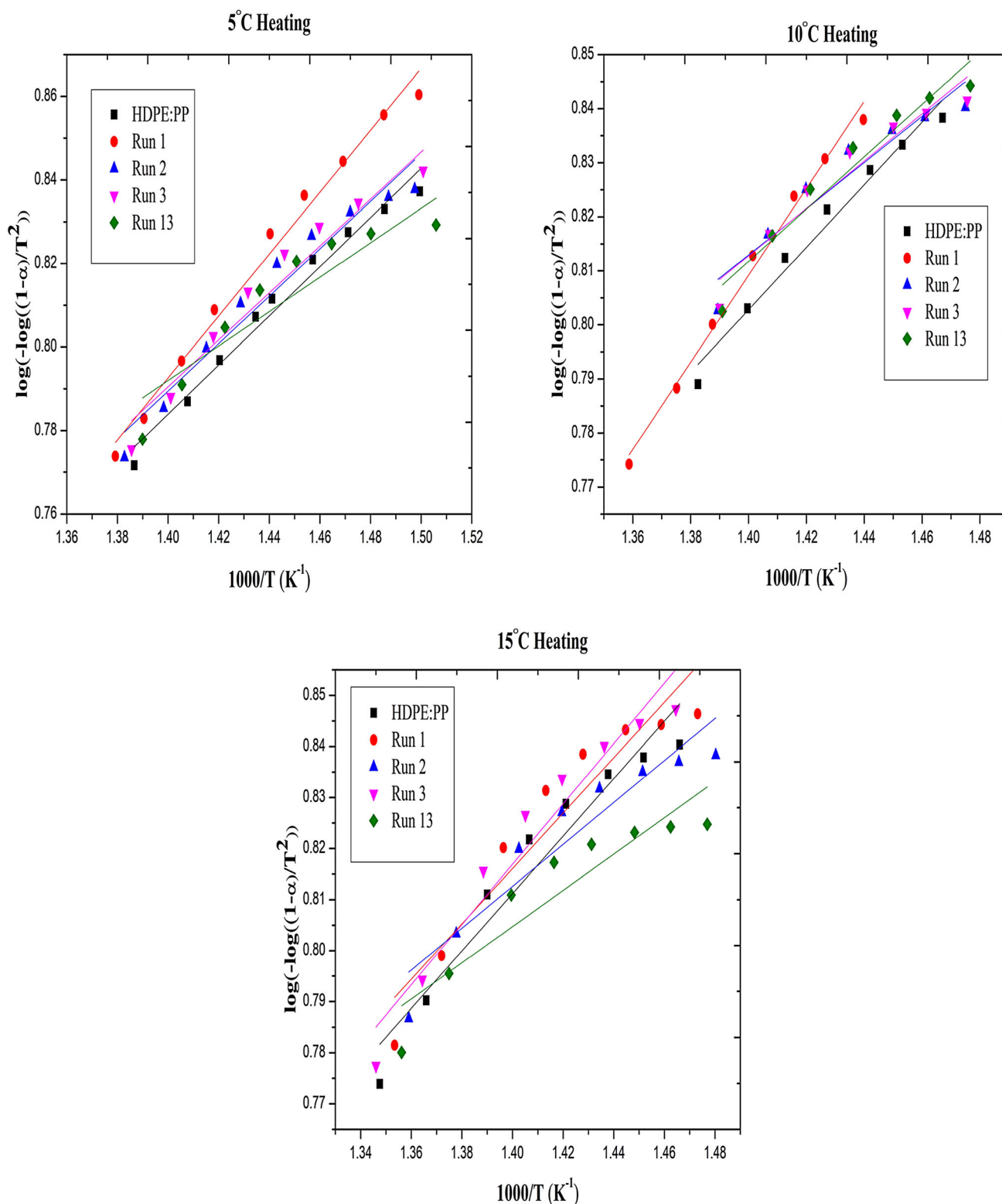


Figure 11. Coats and Redfern plots for activation energy determination of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends.

where Z is the pre-exponential factor, k is the Boltzmann constant, h is the Planck constant, ΔS is a change in activation entropy. T_m is the peak melting temperature. Plot to find E_a is $\ln(\alpha/T^2)$ vs. $1/T$ and the slope $-E_a/R$.

Figure 10 shows the Horowitz and Metzger plots for HDPE/PP, Runs 1, 2, 3 and 13. Adding NC to the matrix improved the activation energy of polymer blends. Comparing pure HDPE/PP with blends containing 1.5% NC and

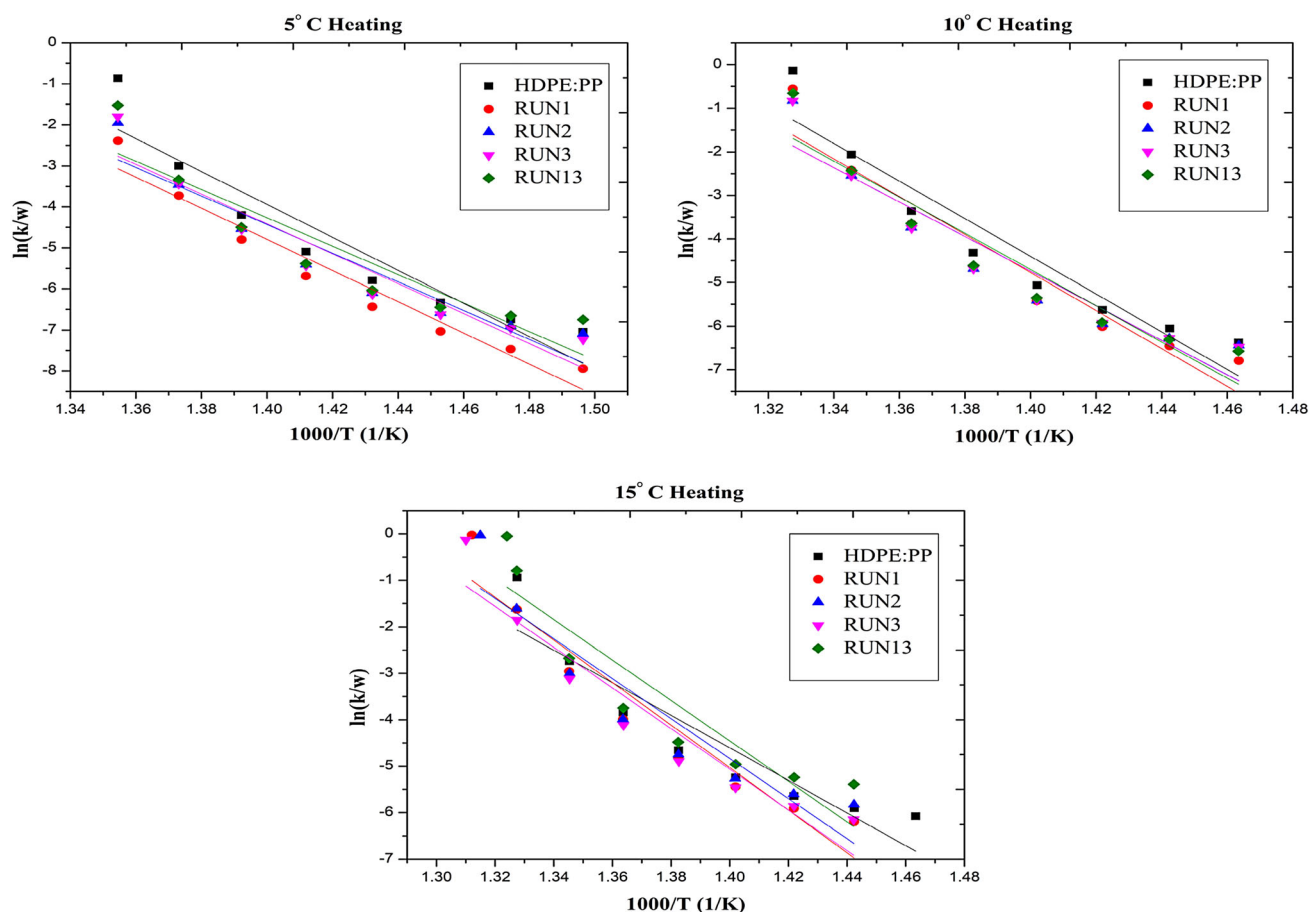


Figure 12. Chan plots for activation energy determination of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends.

3% NC at a rate of heating of 5°C per min, the activation energy of HDPE/PP is 113 kJ mol⁻¹, which increased to 141 kJ mol⁻¹ for the 1.5% NC polymer blend, on further addition of 3% NC, the activation energy of blends reduces to 122 kJ mol⁻¹. The decrease in activation energy with increasing levels of NC addition was slight. It is due to the formation of nano-aggregates, which results in quicker degradation of composites under high temperatures. Figure 11 illustrates the Coats and Redfern models, while figure 12 depicts the Chan plots to predict the activation energy. Both the models predicted activation energy values lower than the predictions made by the Horowitz and Metzger models. The activation energy values for HDPE/PP, Run 1 and Run 2 at a 5°C per min heating rate for the Coats model was 82.5, 84.4 and 82.9 kJ mol⁻¹, and for the Chan model, they are 29.6, 26.7 and 29.6 kJ mol⁻¹, respectively. The higher heating rate increases the activation energies of polymer blend samples. Figure 13 shows Broida plots and Figure 14 provides the Piloyan-Novikova plots for activation energy determination. The observed values of activation energy for HDPE/PP, Run 1 and Run 2 at a 5°C per minute heating rate for the Broida model is

90.7, 86.2 and 81.2 kJ mol⁻¹, respectively, for the Piloyan-Novikova model, they are 52.8, 53 and 53 kJ mol⁻¹, respectively. Each additive added significantly improves the activation energies in runs, and in combination, both NC and pEMAH result in better thermal stability. The activation energy increases with higher heating rates irrespective of levels of NC addition to the matrix. While each model sheds light on unique aspects regarding variations of composition with activation energy at different heating rates, one common observation among all models prevails: an increase in NC content leads to higher activation energy (except for some specific cases), which further supports the relationship between these two parameters. The Horowitz and Metzger model consistently demonstrates the elevation of activation energies for increased levels of NC and pEMAH regardless of the heating rate value. The Coats and Redfern models also depict comparable patterns: the activation energies increase with changes in composition with some differences among rates. Similarly, the Chan, Broida and Piloyan-Novikova models indicate higher activation energy values as NC content rises with the heating rates. It underscores the strength this composite material holds

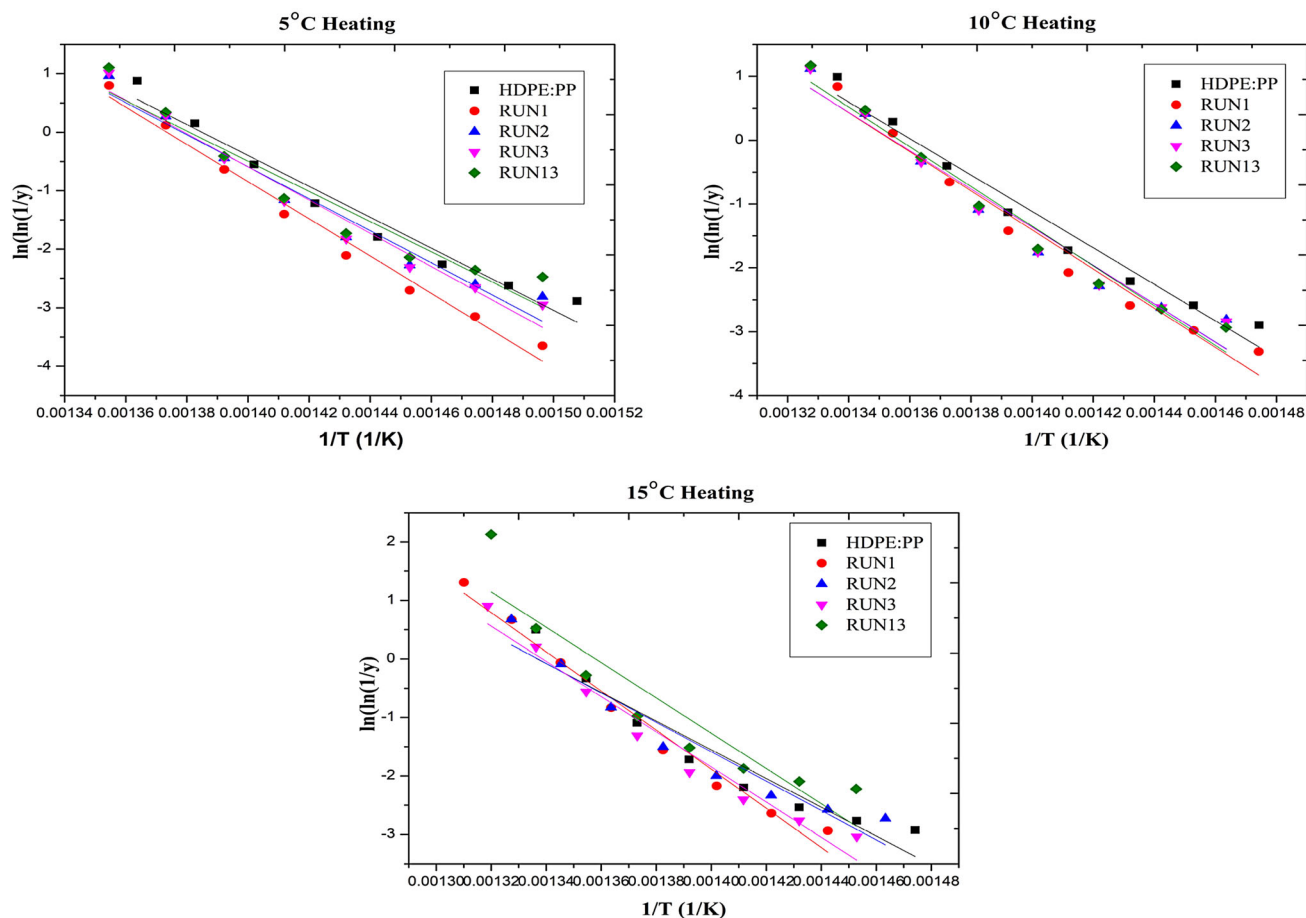


Figure 13. Broida plots for activation energy determination of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends.

against various compositions at different heating rates due to thermal stability.

3.9 X-ray diffraction analysis of NC-dispersed HDPE/PP blends

Figure 15 provides the XRD spectral variations of polymer blends prepared in Runs 2, 3 and 4. The XRD spectra of the synthesized samples ranged between 14° to 24° (2θ values). The HDPE/PP blend diffraction pattern shows a combination of crystalline and amorphous peaks for both HDPE and PP. Specific diffraction peaks for PP are observed at (110), (040) and (130), while for HDPE, they are at (110) and (200). The introduction of NC does not alter peak positions, indicating no effect on the crystal structure. Runs 2, 3, and 4 have the same peak positions, but there is variation in peak intensities. Notably, Runs 2, 3 and 4 show a strong peak at 21.49° and a less intense peak at 23.96° . Adding pEMAH reduces peak intensities in the composite with increased maleic anhydride content and decreased NC content, resulting in a reduced crystalline phase. The peak intensity decreases from 35417 cps in Run 2 to 25637 cps by adding

1.5 wt% of pEMAH (Run 3). Subsequently, it decreases to 24888 cps in Run 4 with reduced NC content.

4. Conclusions

FTIR spectral analysis of absorption in Run 1 indicates the carbon-oxygen asymmetric bond stretch at 1860 cm^{-1} and 1188 cm^{-1} . It represents the existence of carbonyl asymmetric ring stretching and the existence of maleic anhydride. The peak at 415 cm^{-1} shows silicon and aluminium, which indicates NC is present. Run 4 and Run 9 blends have 1.5% and 3% NC. They reach a tensile strength of 28.6 MPa. The specimens exposed to thermal aging create crosslinking characteristics in the polymer structure and improve the tensile strength of the specimens. The Tsai-Pagano model, the Christensen-Waals model, and the ROM model overestimated the modulus of composites in comparison with experimental values. The ratio of the modulus of composites to the matrix varies linearly with the volume fraction of fillers for all five micromechanical models studied. From the contour plot drawn for NC% vs. pEMAH%, we can infer that 3% NC and 1% pEMAH

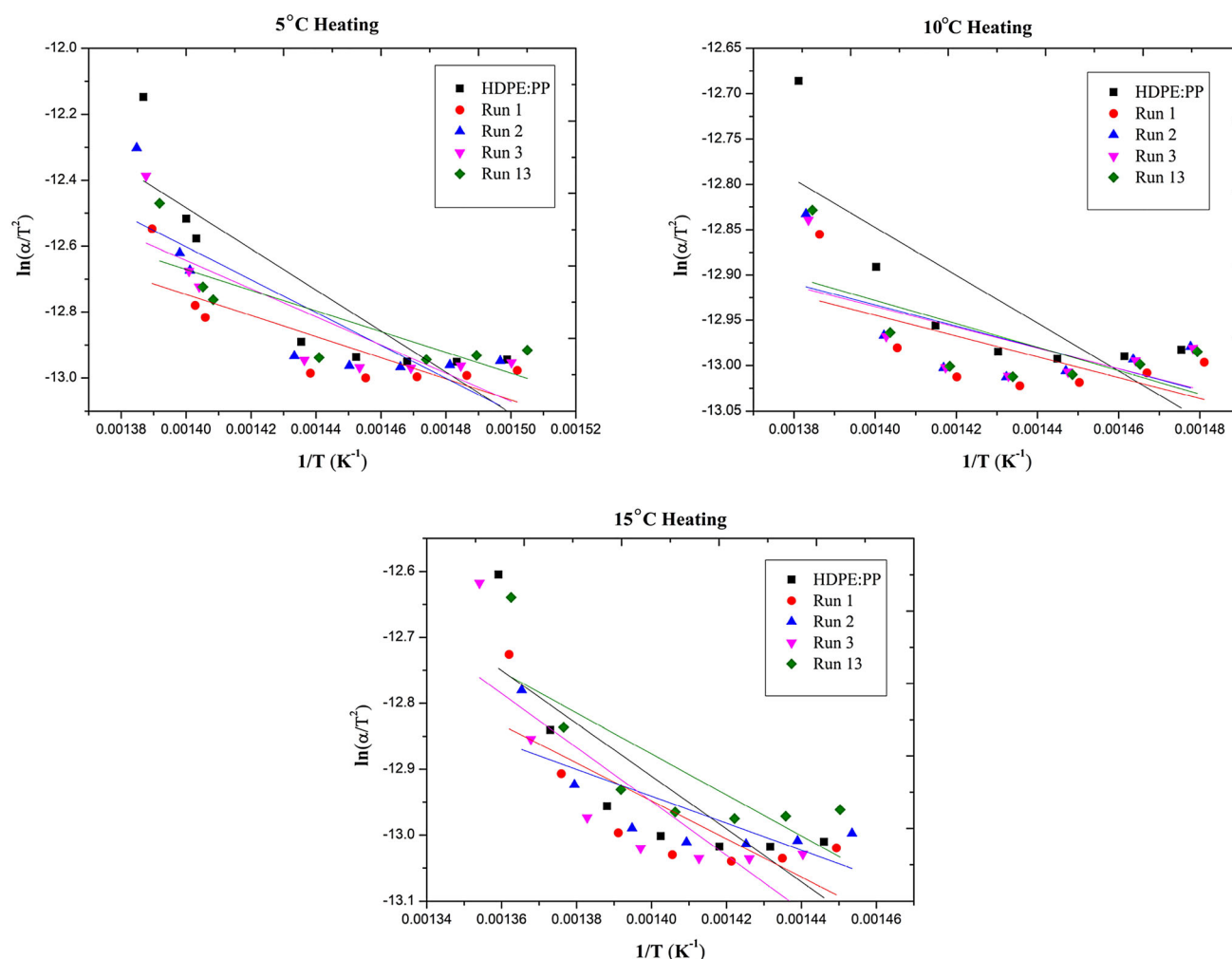


Figure 14. Piloyan-Novikova plots for activation energy determination of HDPE/PP, Run 1, Run 2, Run 3 and Run 13 polymer blends.

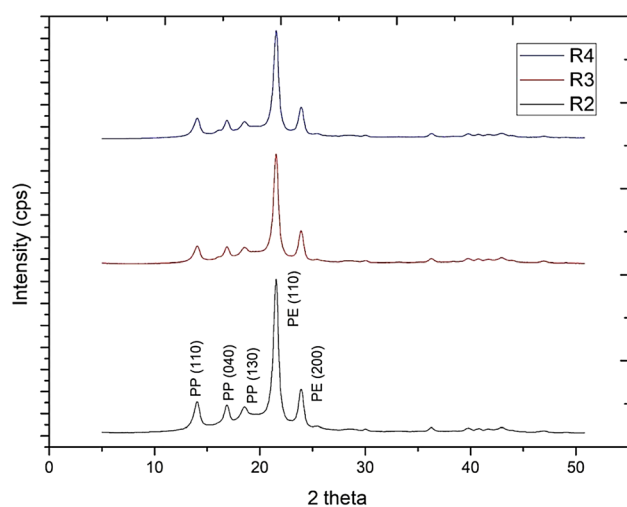


Figure 15. XRD spectra of HDPE/PP/NC blends (Run 2, Run 3 and Run 4).

polymer blends achieve a maximum tensile strength of over 29 MPa. The contour plots show that 3% NC polymer mixes have the best strength. The K_c values of aged samples go up with more filler. K_c values in Run 1 and Run 2 before aging are 1.36 and 1.35 MPa m^{0.5} and rise to 1.54 and 1.57 MPa m^{0.5} after aging. The fracture energy release rate for Run 1 and Run 2 start at 2.08 and 2.35 kJ m⁻² before aging and once aged, they increase to 6.17 and 5.82 kJ m⁻². High heat on tensile samples leads to bonding in the polymer. The bonding makes a polymer firm, prevents bending, and increases the tensile strength. The COMSOL models predict the tensile strength of simulated values at 27 MPa for the 1.5% NC-reinforced matrix, which is same as experimental tensile value.

The HDPE/PP blend has an onset temperature of 403°C at a 15°C rate. Adding 1.5% NC boosts this to 418°C. The onset temperature of blends dispersed with NC is higher than that of the pure polymers, which enhances the stability of the polymers at high temperatures. The comparison of

Runs 2 and 3, containing 3% NC and 0.5% and 1.5% pEMAH, respectively, indicates that the temperature at 50% weight loss values exhibits no significant changes with the addition of compatibilizers. Clay loading into blends improves their resistance to the rise in temperature of polymer blends. Adding NC to polymer blends improved the activation energy of the polymer matrix. The activation energy of pure HDPE/PP blends predicted by the Horowitz and Metzger model is 113 kJ mol^{-1} , and adding 1.5% NC increased the activation energy value to 141 kJ mol^{-1} . The Coats and Redfern method and the Chan models predicted the activation energy less than the Horowitz and Metzger models.

Acknowledgments

The corresponding author thanks the Department of Chemical Engineering, Amrita Vishwa Vidyapeetham, Coimbatore, for the continuous support.

References

- [1] Alshammari B A, Alenad A M, Al-Mubaddel F S, Alharbi A G, Al-Shehri A S, Albalwi H A *et al* 2022 *Polymers* **14** 3427
- [2] Melton George H, Edward N and Peters Ruth K (ed) 2011 *Arisman engineering thermoplastics In: Applied plastics engineering handbook: Processing and materials* (Elsevier) 7–21
- [3] Patil N A, Njuguna J and Balasubramanian K 2020 *Eur. Polym. J* **125** 109529
- [4] Patel K, Chikkali S H and Sivaram S 2020 *Prog. Polym. Sci.* **109** 101290
- [5] Padhy V and Kandasubramanian B 2024 *Polym. Bull.* **81** 15259–15292
- [6] Jose S, Aprem A S, Francis B, Chandy M C, Werner P, Alstaedt V *et al* 2004 *Eur. Polym. J* **40** 2105–2115
- [7] Luo Y, Lin X, Lichtfouse E, Jiang H and Wang C 2023 *Environ. Chem. Lett.* **21** 3127–3158
- [8] Green C, Vaughan A, Stevens G, Pye A, Sutton S, Geussens T and Fairhurst M J 2015 *IEEE Trans. Dielectr. Electr. Insulat.* **22** 639–648
- [9] Coughlin M L, Huang D E, Edgar C M, Kotula A P and Migler K B 2024 *Macromolecules* **57** 9751–9765
- [10] Argoud A, Trouillet-Fonti L, Ceccia S and Sotta P 2014 *Eur. Polym. J* **50** 177–189
- [11] Liu W, Cheng L and Li S 2018 *Composit Commun.* **10** 221–225
- [12] Boxue D, Zhonglei L, Shuofan Z and Mingsheng F 2021 *J. Electr. Eng.* **16** 2
- [13] Jubinville D, Esmizadeh E, Tzoganakis C and Mekonnen T J C P B E 2021 *Compos. B Eng.* **219** 108873
- [14] Adnan M, Abdul-Malek Z, Lau K Y and Tahir M 2021 *IET Nanodielectr.* **4** 84–97
- [15] Gabr M H, Okumura W, Ueda H, Kuriyama W, Uzawa K and Kimpara I 2015 *Compos. B Eng.* **69** 94–100
- [16] Abd El-Baky M A, Attia M A and Kamel M 2018 *Plast. Rubber Compos.* **47** 47–64
- [17] Khalaj M J, Ahmadi H, Lesankhosh R and Khalaj G 2016 *Trends Food Sci. Technol.* **51** 41–48
- [18] Sutar H, Mishra B, Senapati P, Murmu R and Sahu D 2021 *J. Compos. Sci.* **5** 24
- [19] Abuoudah C K, Greish Y E, Abu-Jdayil B, El-said E M and Iqbal M Z 2021 *J. Appl. Polym. Sci.* **138** 50024
- [20] Wiphanurat C, Hanthanon P, Ouipanich S, Harnkarnsujarit N, Magaraphan R and Nampitch T 2023 *Polym. Bull.* **80** 11421–11437
- [21] Ebadi-Dehaghani H, Khonakdar H A, Barikani M and Jafari S H 2015 *Compos. B Eng.* **69** 133–144
- [22] Li C, Zha J, Long H, Wang S, Zhang D and Dang Z 2017 *Compos. Sci. Technol.* **153** 111–118
- [23] Romina E, Sharif F, Rashedi R and Nejad A D 2022 *J. Appl. Polym. Sci.* **139** 51750
- [24] Malek A, Khaled M and Mohamed D 2019 *J. Nanomater.* **1–8** 4184351
- [25] de A S da Silva G and d’Almeida J R M 2022 *Polym. Compos.* **30** 1–13
- [26] Mohammad H K, Seyed M M M, Saeid A, Hamed A and Rostami-Daroukola M R 2022 *Polym. Compos.* **43** 389–398
- [27] Zhang H-C, Kang B, Chen L-S and Xiang L 2020 *Polym. Test.* **87** 106521
- [28] Sumana M, Dhibar A K and Khatua B B 2010 *J. Appl. Polym. Sci.* **116** 1010–1020
- [29] Abbas G, Negin S J, Seyyed A H, Mohammad A and Mohammadreza N 2020 *Polym. Compos.* **41** 3748–3757
- [30] Kovács T, Simon-Stger L, Heller B and Varga C S 2021 *J. Polym. Res.* **28** 294
- [31] Liu Y, Zhang X, Gao Q, Huang H, Liu Y, Min M *et al* 2021 *Polymers* **13** 1954
- [32] Bujjibabu G, Das V C, Ramakrishna M and Nagarjuna K 2021 *Nano Hybrids Compos.* **32** 85–97
- [33] Andre J S, Li B, Chen X, Paradkar R, Walther B, Feng C *et al* 2020 *Polymer* **212** 123141
- [34] Hongfei C, Zhou Y and Liu Q 2019 *Kaolinite Nanomaterials: Preparation, properties and functional applications In: Nanomaterials from clay minerals: a new approach to green functional materials* 285–334 (China: Elsevier) eBook ISBN: 9780128145340
- [35] Roshan S R, Mahalingam V, Lakshmi Narayanan E R, Naveen Balaji N, Murali B, Krishna Prasad R *et al* 2018 *Mater Res. Exp.* **5** 046204
- [36] Ivankova E, Kasatkin I, Vaganov G, Elokhovskiy V, Bugrov A, Yudin V *et al* 2022 *Materials* **15** 3531
- [37] Mahmoud Mohamed E, Khalifa M A, Youssef M R and El-Sharkawy R M 2022 *J. Appl. Polym. Sci.* **139** 1–17
- [38] Mousavi Fakhrabadi S H, Ahmadi S and Arabi H 2022 *Polym. Degrad. Stability* **198** 109882
- [39] Hashim A, Kadham A J, Algidisawi H A, Hadi A and Habeeb M A 2021 *Nanosistemi, Nanomateriali, Nanotehnologii* **19** 0091–0102
- [40] Habeeb M A and Mohammed A H 2023 *Opt. Quant. Electron* **55** 791
- [41] Hashim A and Habeeb M A 2018 *J. Bionanosci.* **12** 660–663
- [42] Baioumy M G S, Jonathan C, John P and Kyriaki K 2019 *Polym. Eng. Sci.* **59** E330–E338

- [43] Aarathi V, Harshita E, Nalinashan A, Ashok S and Krishna Prasad R 2019 *Int. J. Sustain. Energy* **38** 333–347
- [44] Nanoth R, Karingamanna J, Ramakrishnan P, Ganesan R N, Pannerselvam R and Annusamy V V 2019 *J. Vinyl Addit. Techn.* **25** 1–14
- [45] Abthagir P S and Saraswathi R 2005 *Mater. Chem. Phys.* **92** 21–26
- [46] Lu K M, Lee W J, Chen W H and Lin T C 2013 *Appl. Energ.* **105** 57–65
- [47] Arun Kumar A, Antony Lilly Grace M, Subramaniyan Raja R, Rama S and Pragadeeswari G 2021 *Mater. Today Proc.* **47** 4852–4860

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.